

# Regression: Understanding Relationships Between Variables



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# Regression: Understanding Relationships Between Variables



# Roadmap

- + Introduction to Regression
- + The Regression Line
- + Least Squares Method
- + Interpreting Regression Results
- + Residuals and Residual Plots
- + Assumptions and Limitations
- + Real-World Applications
- + Practice and Hands-on Analysis?



# Introduction to Regression



# What is Regression?

- + So far, we've explored how variables move together using bivariate statistics.
- + However, we haven't considered these variables within a system where we have:
  - + DV (dependent variable): The outcome we're interested in
  - + IV (independent variable): The predictor or factor we think influences the DV.
- + Regression is a useful statistical method that allows us to understand how the DV changes as the IV changes.
  - + It helps in modeling and analyzing relationships between variables.



# Example: Smoking and Lung Capacity

- + Let's revisit our earlier example examining the relationship between smoking and lung capacity.
- + We're interested in how the number of cigarettes smoked affects lung capacity.

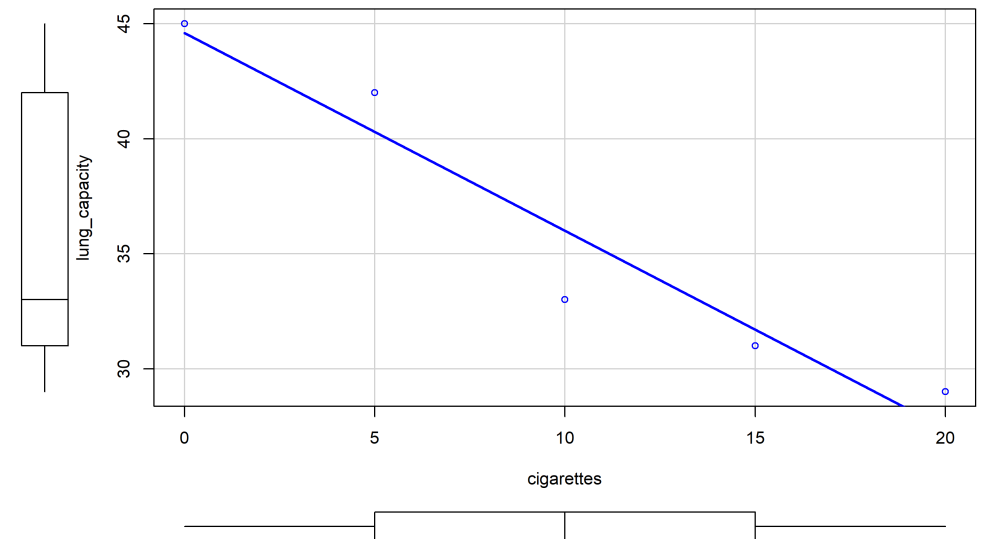
```
# Create the data frame
smoking <- data.frame(
  cigarettes=c(0,5,10,15,20),
  lung_capacity=c(45,42,33,31,29))
```

| cigarettes | lung_capacity |
|------------|---------------|
| 0          | 45            |
| 5          | 42            |
| 10         | 33            |
| 15         | 31            |
| 20         | 29            |



# Interpreting Smoking and Lung Capacity

- + When we plot these data, we observe a **negative linear association between** the number of cigarettes smoked and lung capacity.
- + The **correlation coefficient** between lung capacity and cigarette consumption is  $-0.962$ .
  - + This value indicates a strong negative relationship.
- + The **coefficient of determination** (r-squared) is 0.924, meaning that approximately 92.4 of the variation in lung capacity can be explained by cigarette consumption.
- + Knowing the number of cigarettes someone smokes allows us to predict their lung capacity.
- + The higher the correlation (and by extension, variance explained) the more accurate our predictions *ought* to be.



# The Regression Line





# What is a Regression Line?

- + Simple explanation:
  - + We draw a line through the points in a scatter plot to summarize the relationship.
- + Technical explanation:
  - + We fit a linear model of the relationship between  $x$  and  $y$ .
- + A regression line is a straight line that describes how a response variable  $y$  changes as an explanatory variable  $x$  changes.
- + We often use a regression line to predict the value of  $y$  for a given value of  $x$ ,
  - + when we believe the relationship between  $x$  and  $y$  is linear.

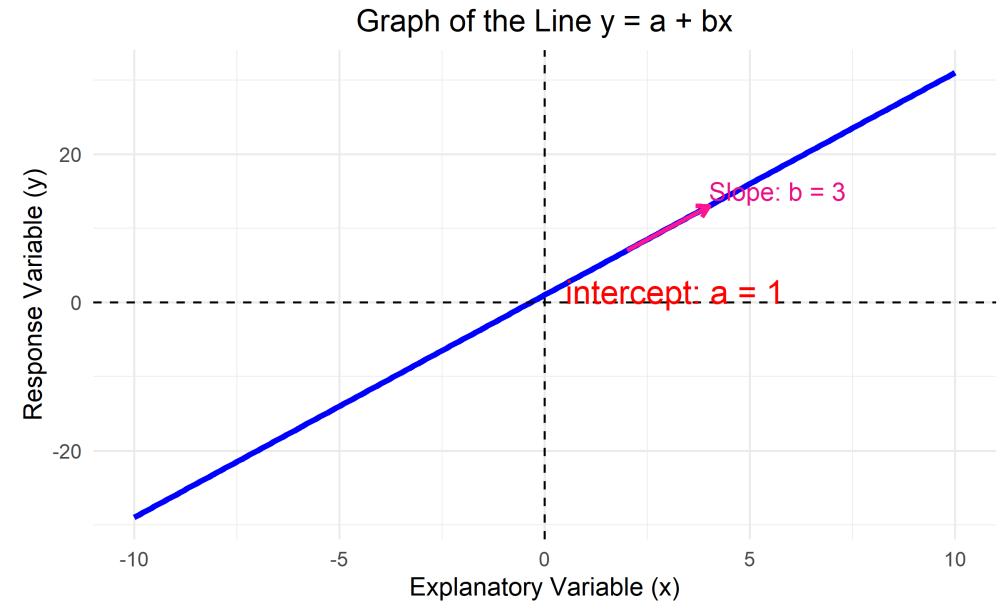


# Equation of a Line

- + We can describe this line in a familiar equation
- + Suppose that  $y$  is a response variable (plotted on the vertical axis) and  $x$  is an explanatory variable (plotted on the horizontal axis).
- + A straight line relating  $x$  to  $y$  has an equation of the form

$$y = a + bx; (\text{or } y = mx + b)$$

- + Example where  $a = 1$  and  $b = 3$



# Components of the Equation

$$y = a + bx; (\text{or } y = mx + b)$$

- + In this equation,
  - +  $b$  ( $m$ ) is the slope:
    - + The amount by which  $y$  changes when  $x$  increases by one unit.
- +  $a$  ( $b$ ) is the intercept:
  - + The value of  $y$  when  $x = 0$ .
- + With those two points, any straight line can be defined
  - + within the Cartesian plane (except lines parallel to the  $y$ -axis).



# Understanding the Slope

- + Positive slope:
  - + As  $x$  increases,  $y$  increases.
- + Negative slope:
  - + As  $x$  increases,  $y$  decreases.
- + The magnitude of the slope indicates the steepness of the line.
  - + A larger absolute value of  $b$  means a steeper line



# Understanding the Intercept

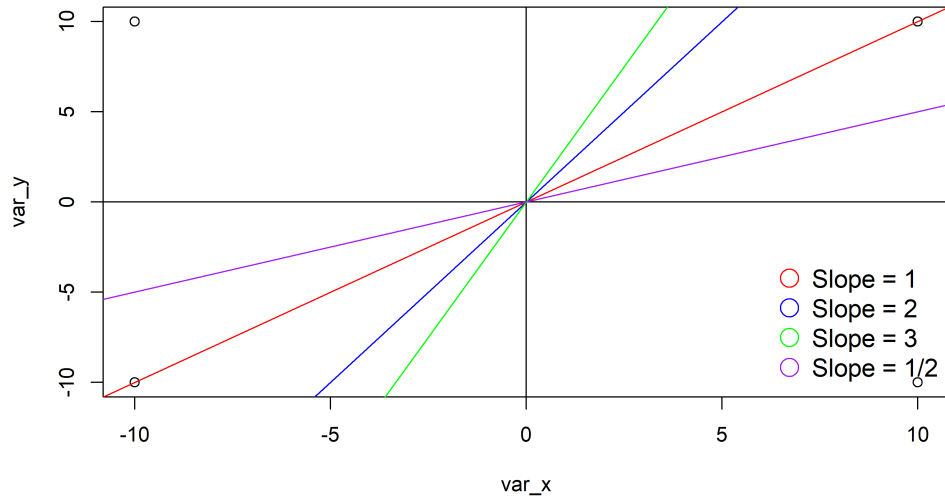
- + The intercept ( $a$ ) represents the starting value of  $y$  when  $x = 0$ .
- + It determines where the line crosses the  $y$ -axis.
- + Changing the intercept shifts the line up or down without altering its slope.



# Visualizing Different Slopes

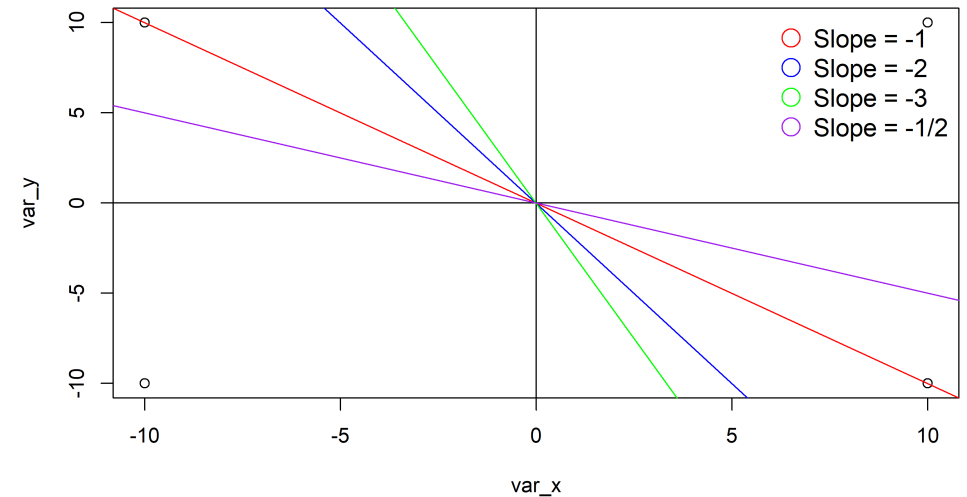
+ Here, we demonstrate lines with different positive slopes.

+ A higher slope means a steeper line.



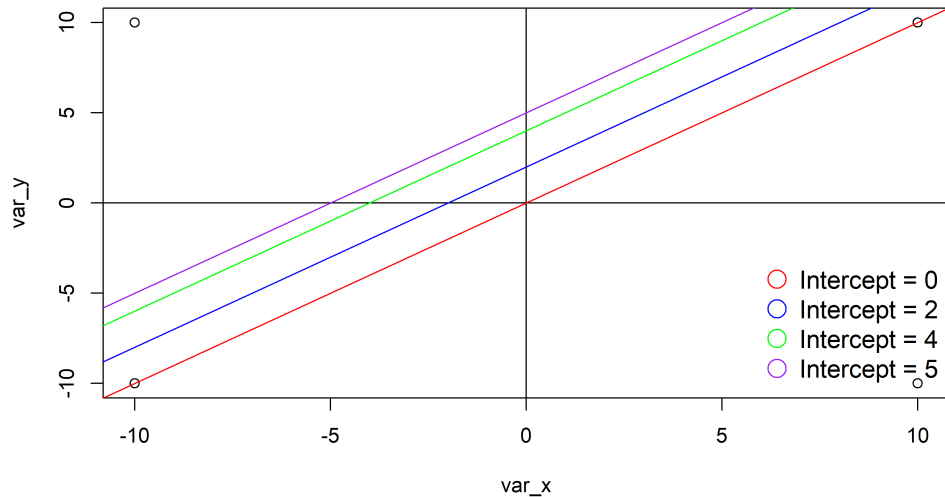
+ In this plot, we demonstrate lines with different negative slopes.

+ Negative slopes show a downward trend as  $x$  increases.

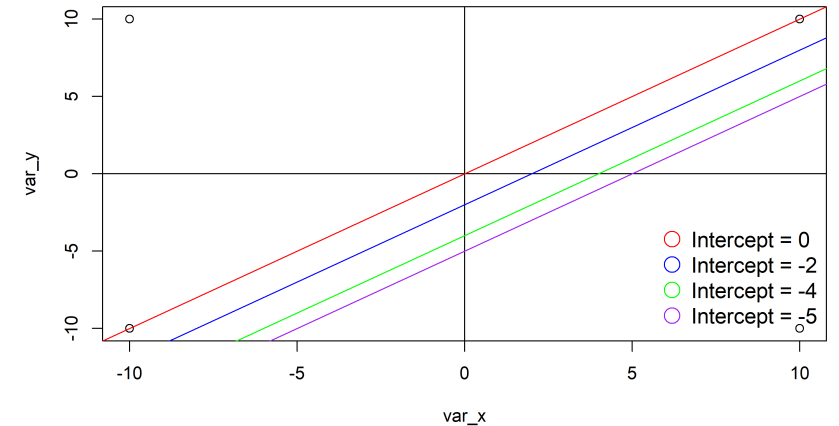


# Visualizing Different Intercepts

- + This plot demonstrates how changes in the intercept shift the line vertically.
- + A higher intercept means the line starts higher on the y-axis.



- + In this plot, we explore lines with negative intercepts, shifting the line down on the y-axis.
- + A more negative intercept places the line lower on the y-axis.



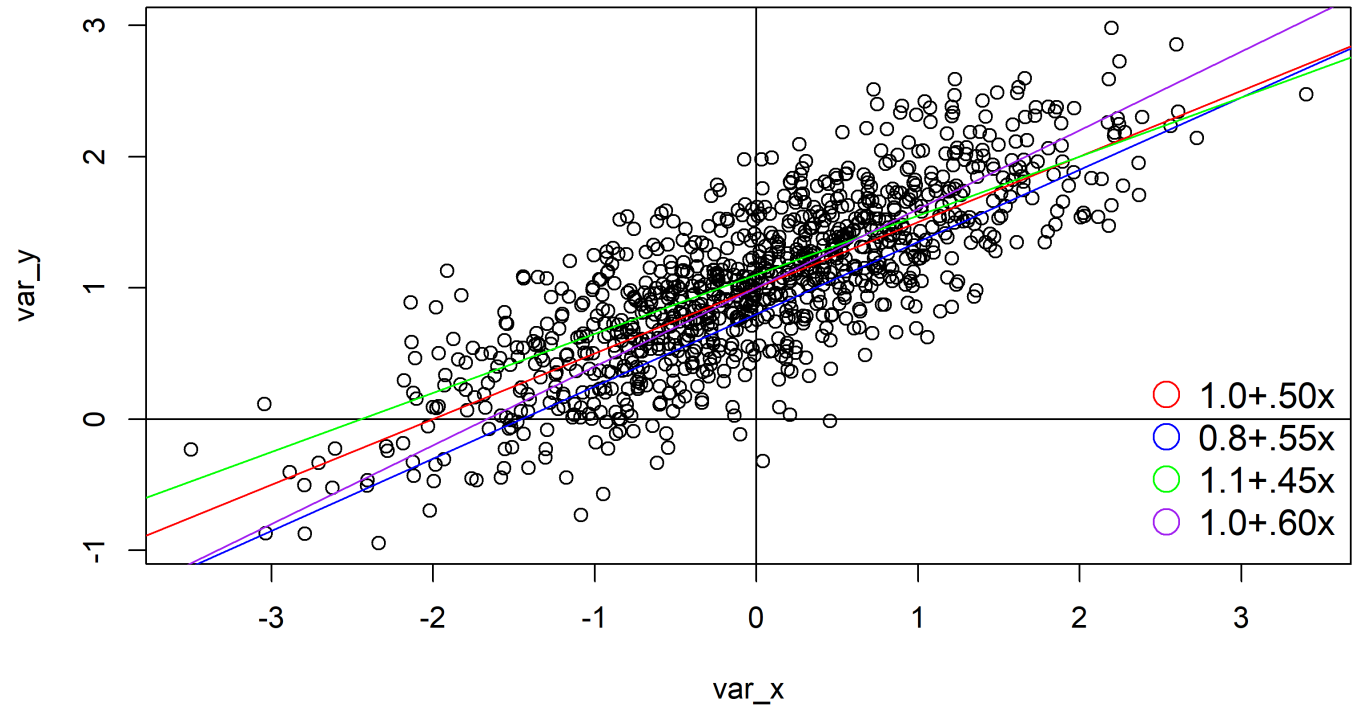
# Least Squares Method





# Objective: Find the "best-fitting" line

- + When we have a set of data points, we often want to find a line that **best fits** the data.
- + To eliminate the subjectivity of creating a line to fit the data,
  - + we need an objective way to draw the line.
- + The least squares method is the most popular approach.



# Goal of Least Squares

- + The least-squares regression line of  $y$  on  $x$  is the line that makes the **sum of the squares** of the vertical distances of the data points from the line as small as possible.
- + **Mathematically**, we aim to:

$$\text{minimize } \sum_{i=1}^n (y_i - (a + bx_i))^2$$

- + Sometimes the principle of least squares is described as minimizing the sum of the:
  - + squares,
  - + squared residuals, or
  - + squared errors.



# In other words...

- + The least-squares regression line of  $y$  on  $x$  is the line that makes
  - + the sum of the squares of the vertical distances of the data points from the line as small as possible.
- + The goal is to minimize the difference between actual and predicted values of the dependent variable  $y$ 
  - +  $\min\left(\sum_{i=1}^n (y_i - (a + bx_i))^2\right)$
  - +  $\min\left(\sum_{i=1}^n (y_i - \hat{y}_i)^2\right)$
  - +  $\min\left(\sum_{i=1}^n (e_i)^2\right)$
- + This method ensures that the total error between the data points and the regression line is as small as possible.



## Why squares?

- + Squaring emphasizes larger errors and avoids negative residuals canceling out positive ones.



# Mathematical Formulation

- + We have data on an explanatory variable  $x$  and a response variable  $y$  for  $n$  individuals.
- + From the data, calculate the means  $\bar{x}$  and  $\bar{y}$ 
  - + and the standard deviations  $s_x$  and  $s_y$  of the two variables and
  - + their correlation  $r$ .
- + The least-squares regression line is the line:

$$\hat{y} = a + bx$$

- + with slope:

$$b = r \frac{s_y}{s_x}$$

- + and intercept:

$$a = \bar{y} - b\bar{x}$$

- + The least-squares regression line always passes through  $(\bar{x}, \bar{y})$  and  $(0, a)$  on the graph of  $y$  against  $x$ .



# Interpreting Regression Results



# Calculating Regression Line for Smoking Data

- + Using our smoking data, we can estimate the slope and intercept
- + **Compute the necessary statistics:**

```
# Means
mean_x <- mean(smoking$cigarettes)
mean_y <- mean(smoking$lung_capacity)

# Standard deviations
sd_x <- sd(smoking$cigarettes)
sd_y <- sd(smoking$lung_capacity)

# Correlation
r_xy <- cor(smoking$cigarettes, smoking$lung_capacity)
```



# Calculating Regression Line for Smoking Data

+ Calculate slope (b):

$$\begin{aligned} b &= r \left( \frac{s_y}{s_x} \right) \\ &= r_{\text{lung capacity, cigarettes}} \frac{s_{\text{lung capacity}}}{s_{\text{cigarettes}}} \\ &= -0.962 \times \frac{7.071}{7.906} \\ &= -0.962 \times 0.894 \\ &= -0.86 \end{aligned}$$





# Calculating Regression Line for Smoking Data: Intercept

+ Calculate intercept (a):

$$\begin{aligned} a &= \bar{y} - b\bar{x} \\ &= \text{lungc}\bar{\text{apacity}} - b\text{ciga}\bar{\text{r}}\text{ettes} \\ &= \text{lungc}\bar{\text{apacity}} - -0.86 * \text{ciga}\bar{\text{r}}\text{ettes} \\ &= 36 - -0.86 \times 10 \\ &= 44.6 \end{aligned}$$



# Combined Regression Equation

- + The **regression equation** for our data is:

$$\hat{y} = a + bx = 44.6 + -0.86x$$

- + Interpretation

- + **Intercept (a)**: Predicted lung capacity when no cigarettes are smoked 44.6
- + **Slope (b)**: For each additional cigarette smoked, we would expect a 0.86 liter decline in lung capacity



# Interpreting the Regression Equation

- + The intercept makes a prediction for the  $y$  outcome, when  $x$  is 0.
- + Here, that means that the expected/predicted lung capacity for a non-smoker is 44.6
- + The slope gives us the predicted change in outcome for a 1-unit increase in  $x$ .
- + For every 1 additional cigarette, we would expect a -0.86 decline in lung capacity



# Using the Regression Equation to Make Predictions

+ **Predict lung capacity for a 5-cigarette smoker:**

+ If we want to predict lung capacity for a 5 cigarette smoker, we use the regression equation to predict  $y$ .

$$\begin{aligned}\hat{y} &= 44.6 + -0.86x \\ &= 44.6 + -0.86 * 5 \\ &= 44.6 + -4.3 \\ &= 40.3\end{aligned}$$

+ Given our equation, we would predict that a 5-cigarette smoker would have a lung capacity of 40.3 liters.



# Two Regression Lines

- + The distinction between explanatory variables ( $x$ ) and response variables ( $y$ ) is essential in regression.

$$\hat{y} = a + b_{yx}x$$

$$\hat{x} = a + b_{xy}y$$

$$b_{yx} \neq b_{xy}$$

- + These  $b$  coefficients are *not* the same
  - + The equation is not symmetric
  - + The slope of the regression line for predicting  $y$  from  $x$  is not the same as the slope of the regression line for predicting  $x$  from  $y$ .



# Relationship Between Regression and Correlation

- + There is a close connection between **correlation** and the **slope** of the least-squares regression line.
- + The **slope** is calculated using the correlation coefficient:

$$b = r \left( \frac{s_y}{s_x} \right)$$

## + Key Points:

- + The **slope** ( $b$ ) and the **correlation coefficient** ( $r$ ) always have the **same sign**.
- + A positive correlation results in a positive slope.
- + A negative correlation results in a negative slope.
- + The magnitude of the slope depends on both the correlation and the standard deviations of  $x$  and  $y$ .



# Standardizing Variables

+ If we **standardize**  $x$  and  $y$  (convert them to z-scores):

+ **Standardized variables:**

$$(z_x = \frac{x - \bar{x}}{s_x})$$

$$(z_y = \frac{y - \bar{y}}{s_y})$$

+ The regression equation becomes:

$$\hat{z}_y = r z_x$$

+ **Interpretation:**

+ The slope of the regression line of standardized variables is the **correlation coefficient** ( $r$ ).

+ The intercept is **zero** because standardized variables have a mean of zero.



# Geometric Interpretation

- + **Standardizing** shifts the origin to the mean and the  $x$  and  $y$  axis are stretched so that  $sd = 1$
- + In the standardized coordinate system:
  - + The regression line passes through the origin  $(0,0)$ .
  - + The slope is equal to the correlation coefficient  $(r)$ .
  - +  $\widehat{z}_y = r_{xy}$





# Implications

- + The correlation coefficient not only measures the strength and direction of a linear relationship but also directly influences the slope of the regression line.
- + Understanding this relationship helps in interpreting the regression results and in assessing how changes in  $x$  are associated with changes in  $y$ .
  - + Along the regression line, a change of 1 standard deviation in  $x$  corresponds to an  $r$  change of standard deviations in  $y$



# Residuals and Residual Plots



# Residual Calculation

- + A residual is the difference between an observed value of the response variable and the value predicted by the regression line.
- + That is, a residual is the prediction error that remains after we have chosen the regression line:
  - + residual = observed  $y$  - predicted  $y$
  - + residual =  $y - \hat{y}$
- + Residuals represent "leftover" variation in the response after fitting the regression line.- Calculated for each data point, residuals help assess the fit of the regression model.
- + The residuals from the least-squares line have a special property:
  - + The **mean** of the least-squares residuals is always **zero**.



# Residual Calculation

```
# Recall
(smoke_regression=lm(lung_capacity~cigarettes,data=smoking))
```

```
##
## Call:
## lm(formula = lung_capacity ~ cigarettes, data = smoking)
##
## Coefficients:
## (Intercept)    cigarettes
##      44.60         -0.86
```

```
prediction <- data.frame(cigarettes = 5)

# Predicted Lung Capacity for 5 Cigarettes
(yhat=predict(smoke_regression,prediction))
```

```
##      1
## 40.3
```

```
# Actual Lung Capacity for 5
(yact= smoking$lung_capacity[smoking$cigarettes==5])
```

```
## [1] 42
```

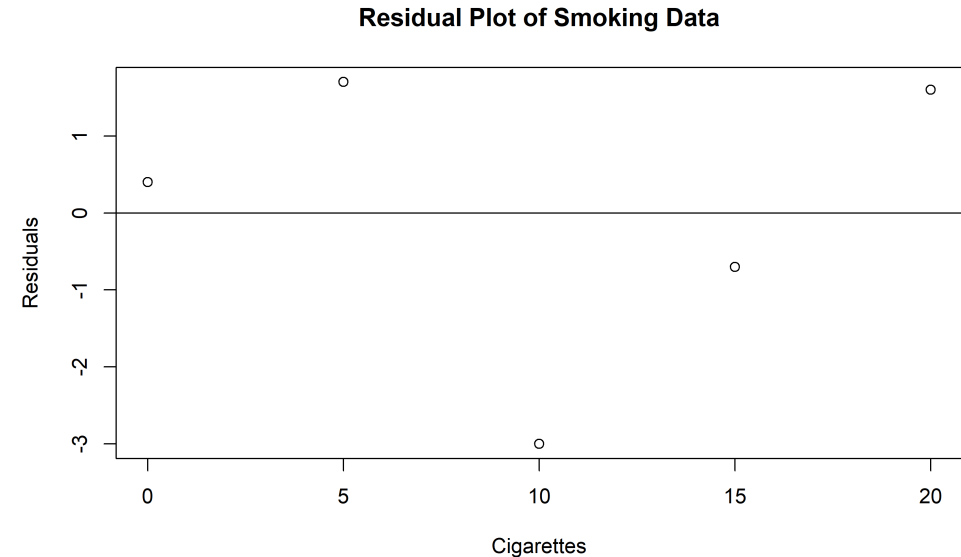
```
#Difference is the Residual
yact-yhat
```

```
##      1
## 1.7
```

```
# Get the Residuals for all the values
(smoke_regression$resid=resid(smoke_regression))
```

```
##      1      2      3      4      5
## 0.4    1.7   -3.0   -0.7    1.6
```

```
# Plotting
plot(smoking$cigarettes, smoke_regression$resid,
     ylab="Residuals", xlab="Cigarettes",
     main="Residual Plot of Smoking Data")
abline(0, 0)
```



# Verification with Actual Data

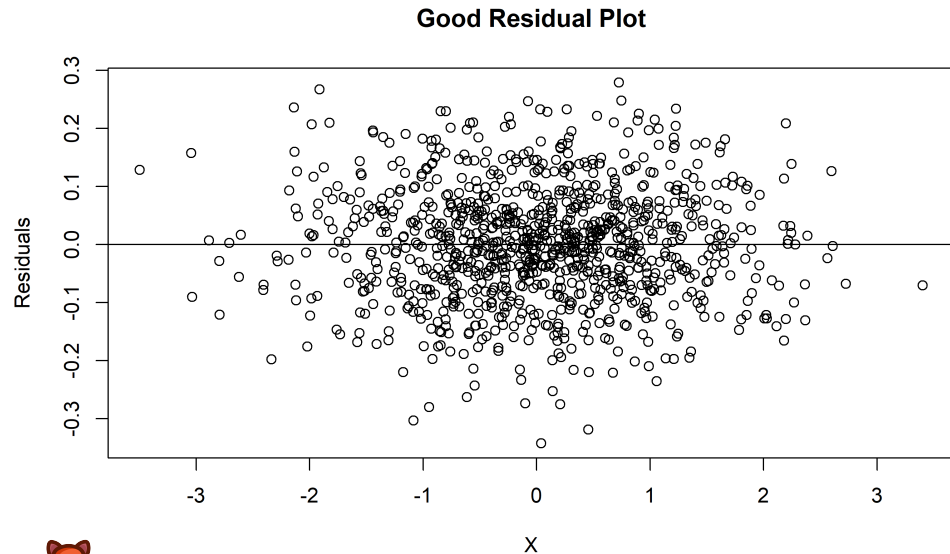
+ Let's compare the predicted values with actual data:

| Actual vs. Predicted Lung Capacity |               |                         |           |
|------------------------------------|---------------|-------------------------|-----------|
| cigarettes                         | lung_capacity | predicted_lung_capacity | residuals |
| 0                                  | 45            | 44.6                    | 0.4       |
| 5                                  | 42            | 40.3                    | 1.7       |
| 10                                 | 33            | 36.0                    | -3.0      |
| 15                                 | 31            | 31.7                    | -0.7      |
| 20                                 | 29            | 27.4                    | 1.6       |

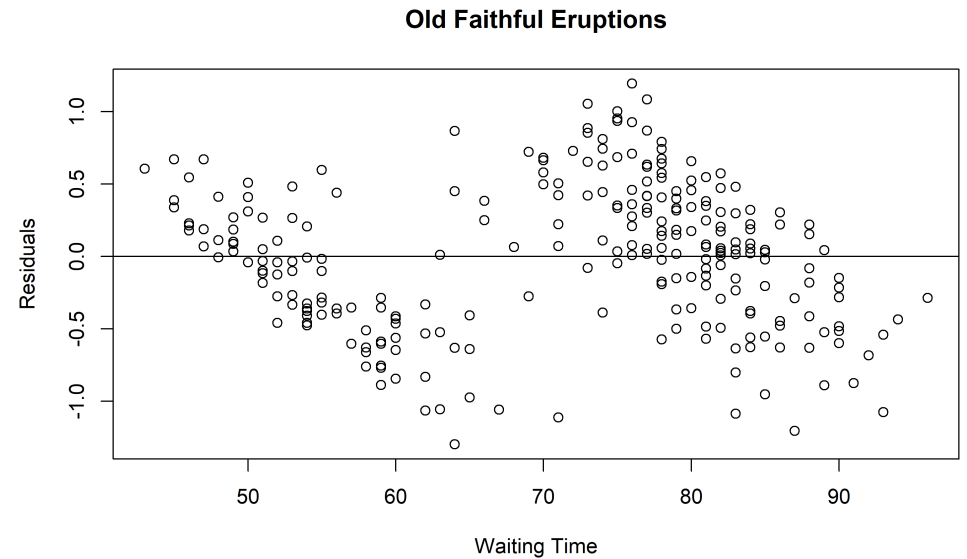


# Residual Plots

- + Plot the residual  $y - \hat{y}$  against the  $x$  value
- + A good residual plot, creates a flat line
- + The residuals are randomly scattered around the line  $y = 0$



- + A concerning residual plot, shows a relationship



# Assumptions and Limitations



# Assumptions of Linear Regression

## 1. Linearity:

- + The relationship between the independent and dependent variables is linear.

## 2. Independence:

- + Observations are independent of each other.

## 3. Homoscedasticity:

- + The residuals have constant variance at every level of  $x$ .

## 4. Normality of Residuals:

- + The residuals are normally distributed.

+ **Violations** of these assumptions can affect the validity of the regression results.





# Alternative Line Estimates

## + **MAD Regression:**

- + Minimizes the **Median Absolute Deviations:**

$$\min\left(\sum_{i=1}^n |y_i - \hat{y}_i|\right)$$

## + **LMS (Least Median of Squares):**

- + Minimizes the median of the squared residuals.

## + **Ridge Regression:**

- + Adds a penalty term to the least squares to prevent overfitting.

## + **Maximum Likelihood Methods:**

- + Estimates parameters by maximizing the likelihood function.



# Real-World Applications



# Beyond Two Variables

- + The residual plot may suggest that other variables are at play.
- + **Multiple Regression:**
  - + Regression can be used to predict  $y$  from multiple  $x$  variables.
  - + Allows for modeling more complex relationships.
  - + Example:
    - + Predicting lung capacity based on cigarettes smoked, age, and exercise level.
- + While multiple regression is powerful, it is beyond the scope of our class.



# Next time is a hands-on lab using...



# Excel

